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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (CST)(CBCS) Degree Examinations
January 2023

22CST113 - ADVANCED DESIGN OF RCC STRUCTURES

Duration: 3 Hours

Max. Marks:100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of **IS:456-2000** is permitted. Assume any missing data suitably.

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the assumptions made in yield line analysis of slabs	5	L*1	1	1
	b) State the characteristic features in yield line analysis of slabs	5	L2	1	1
	c) A hexagonal slab of side 3.6m is simply supported along edges and is isotropically reinforced with 10mm bars at 150mm centre to centre. The effective depth is 125mm and overall depth of slab is 150mm. If M20 concrete and Fe-415 steel are used, determine its load carrying capacity.	10	L3	1	1,2
2.	A flat slab of size 5m x 5m is to be designed without providing drop and column head. Size of columns is 500 x 500 mm and live load on the panel is 4kN/m ² . Take floor finishing load as 1kN/m ² . Use M20 Concrete and Fe-415 steel. Design interior panel of flat slab.	20	L4	1	1,2
3.	A reinforced concrete grid floor is to be designed to cover a floor area of size 12 m x 18 m. The spacing of the ribs in mutually perpendicular directions is 1.5 m c/c. Live load =3 kN/m ² . Adopt M20 Concrete and Fe 415 steel. Assume ends are simply supported. Analyze the grid floors for moments and shears using IS: 456-2000 method and design the suitable reinforcements in the grid floor.	20	L4	2	1,2
Unit – II					
4.	A silo with internal diameter 5.5m, height of cylindrical portion 18m and central opening with 0.5m is to be built to store wheat. Design the silo using M20 concrete and Fe415 steel. Take unit weight of wheat = 8.5kN/m ³ . Angle of internal friction is 28°, Angle of wall friction is 0.75 φ while filling and 0.6 φ while emptying, pressure ratio is 0.5 while filling.	20	L4	3	1,2
5.	Design the side wall and hopper bottom of a rectangular bunker of capacity 300kN to store coal using M20 Concrete and Fe 415 steel. Unit weight of coal is 8 kN/m ³ , angle of repose of coal is 25°.	20	L4	3	1,2
6.	Two span continuous beam of effective span 6m is simply supported at supports. The beam is to be designed to carry a dead load of 16kN/m (inclusive self weight) and a maximum live load of 24kN/m. Allowing 15 percent moment redistribution design the beam of cross section 230 x 500 mm with an effective cover of 50mm. Use M20 concrete and Fe-415 steel. Sketch the reinforcement details.	20	L4	4	1,2

P.T.O.

Unit – III

7. Design a semicircular beam supported on 3 columns equally spaced. The centre of the columns is on a curve of diameter 8m. The superimposed load on the beam per meter length is 20kN. Radius of the girder=4m. Depth of the girder=800mm. Use M20 concrete and Fe 415 steel.

20 L4 5 1,2

8. A circular R.C girder for the foundation raft of a water tower has a mean diameter of 10m. The uniformly distributed load transmitted by eight symmetrically placed columns on the girder being 300kN/m. The width of the beam is 500mm and overall depth is 1000mm. Use M20 Concrete and Fe415 steel. Design the circular girder and sketch the reinforcement details.

20 L4 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
